# Waves Lec 4

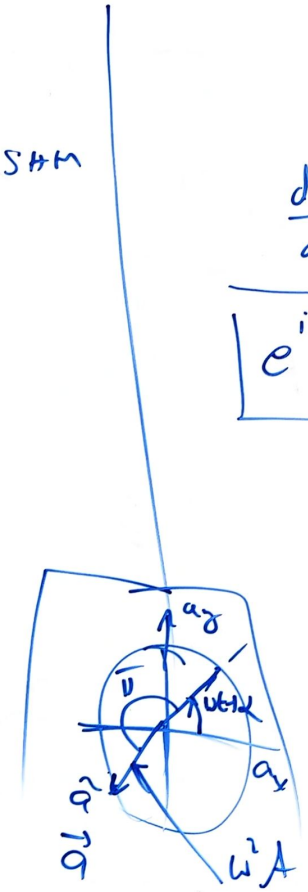
Phasor method of solving SHM from 2nd order ODE

$$\frac{d^2x}{dt^2} = -\omega^2 x, \quad \frac{d^2y}{dt^2} = -\omega^2 y$$

$$\frac{d^2z}{dt^2} = -\omega^2 z$$

$$\tilde{v} = \frac{dx}{dt} + i\frac{dy}{dt}$$

$$\frac{d\tilde{v}}{dt} = \frac{d^2x}{dt^2} + i\frac{d^2y}{dt^2} = -\omega^2 z$$




$$z = A e^{i(\omega t+\alpha)}$$

$$\frac{dz}{dt} = i\omega A e^{i(\omega t+\alpha)}$$

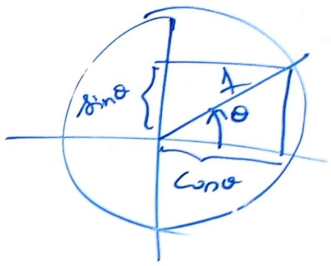
$$e^{i\pi/2} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$$

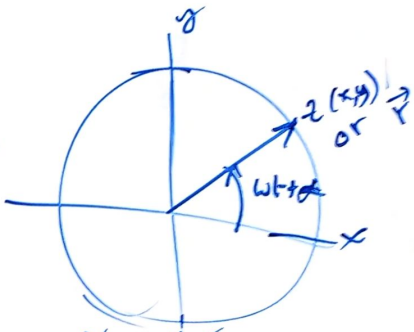
$$\frac{dz}{dt} = \omega A e^{i(\omega t+\alpha+\frac{\pi}{2})}$$

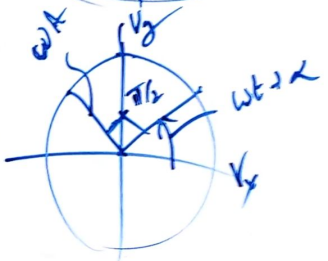
$$\frac{d^2z}{dt^2} = -\omega^2 A e^{i(\omega t+\alpha)}$$

$$-1 = e^{i\pi}$$

$$= \omega^2 A e^{i(\omega t+\alpha+\pi)}$$



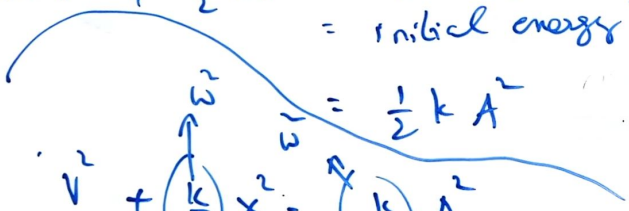



## Energy method

$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{const} = E_0$

$= \text{initial energy}$

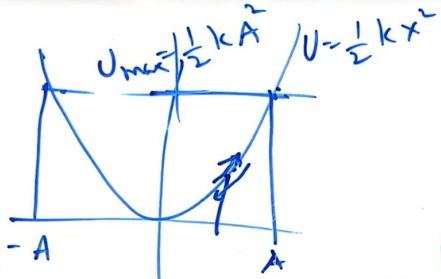
$= \frac{1}{2}kA^2$

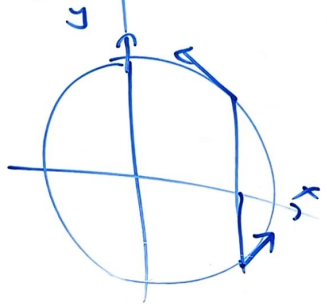


$v^2 + \left(\frac{k}{m}\right)x^2 = \left(\frac{k}{m}\right)A^2$

$v^2 = \left(\frac{dx}{dt}\right)^2 = \omega^2(A^2 - x^2)$

$\frac{dx}{dt} = \pm\omega\sqrt{A^2 - x^2}$




$\frac{dx}{dt} = \omega\sqrt{A^2 - x^2}$

$\int \frac{dx}{\sqrt{A^2 - x^2}} = \omega\int dt$

$x = A\sin\theta$

$dx = A\cos\theta\, d\theta$

$\sqrt{A^2 - x^2} = \sqrt{A^2\cos^2\theta} = A\cos\theta$

$\int d\theta = \omega\int dt$

$\theta = \omega t + C$

$x = A\sin\theta$

$\sin\theta = \frac{x}{A}$

$\theta = \sin^{-1}\frac{x}{A}$

$x = A\sin\theta$

$x = A\sin(\omega t + C)$

$t = 0,\ x(0) = -A$

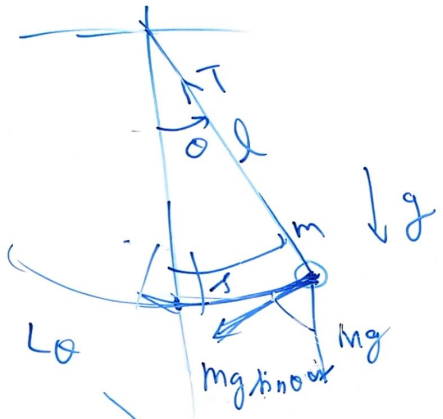
$-A = A\sin C$

$\sin C = -1 \Rightarrow C = -\frac{\pi}{2}$

$x = A\sin(\omega t - \pi/2) = -A\cos\omega t$

# Waves Lec 4

## Simple pendulum




$$m\frac{d^2s}{dt^2} = mL\frac{d^2\theta}{dt^2} = -mg\sin\theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$$

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\left(\theta - \frac{\theta^3}{3!} + \cdots\right)$$

Non-linear

For $\theta \ll \pi/2$

$\sin\theta \approx \theta$

$\approx$

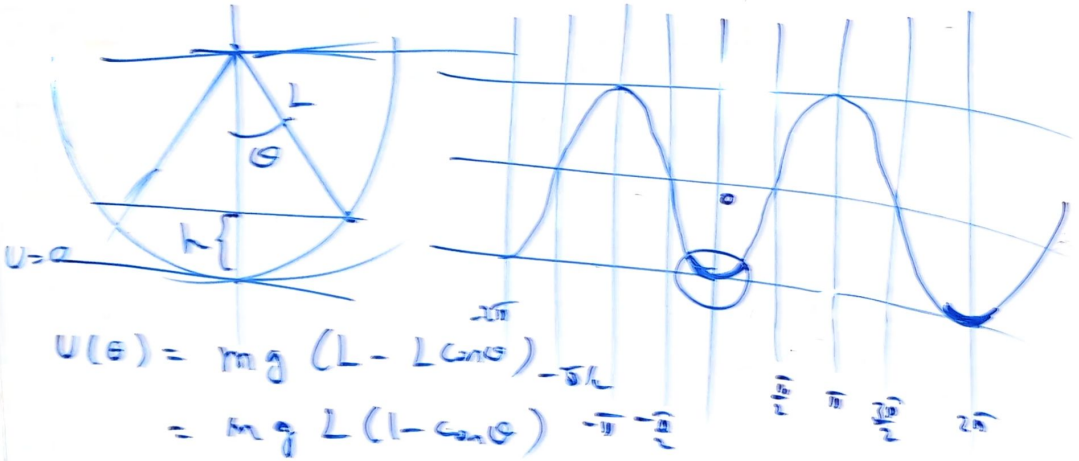
$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\Rightarrow \frac{g}{L} = \omega^2, \quad \omega = \sqrt{\frac{g}{L}}$$

$$\nu = \frac{1}{2\pi}\sqrt{\frac{g}{L}}, \quad T = 2\pi\sqrt{\frac{L}{g}}$$

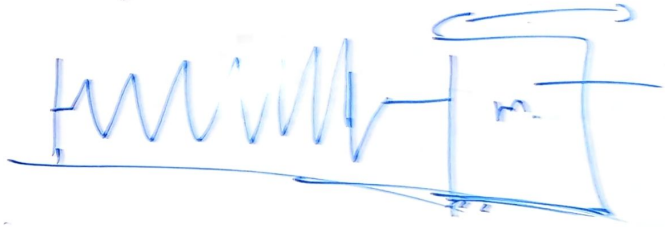
$$\theta(t) = \theta_0 \cos(\omega t + \alpha)$$

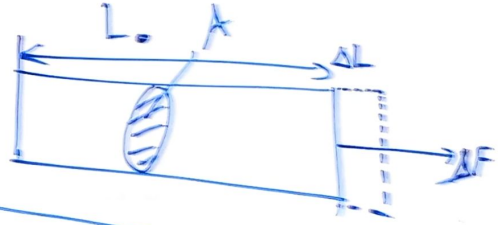



$$U(\theta) = mg(L - L\cos\theta)$$
$$= mgL(1 - \cos\theta)$$

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots$$

$$1 - \cos\theta = \frac{1}{2}\theta^2 - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots$$
$$\approx \frac{1}{2}\theta^2$$




$$\text{strain} = \epsilon = \frac{\Delta L}{L_0}$$

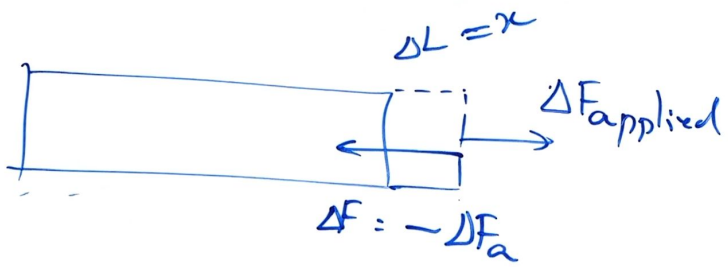
$$\text{Stress} = \frac{\Delta F}{A} = \sigma$$

$$\sigma = Y\epsilon$$

$$\frac{\sigma}{\epsilon} = Y = \text{Young's modulus}$$
- material const.

29-6-16

# Waves Lec 4




$\frac{\sigma}{\epsilon}$

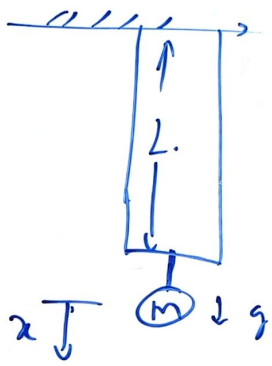
$$\frac{\Delta F_a}{A\epsilon} = Y$$

$$\Delta F = -\frac{AY}{L_0} x$$

$$\Delta F_a = AY \cdot \frac{\Delta L}{L_0}$$

$$= \frac{AY}{L_0} \Delta L$$

$$= \frac{AY}{L_0} x$$




$$m \frac{d^2x}{dt^2} = -\frac{AY}{L_0} x$$

$$\frac{d^2x}{dt^2} = -\frac{AY}{mL_0} x$$

$$= -\omega^2 x$$

$$\omega = \sqrt{\frac{AY}{mL_0}}, \quad Y_{steel} \approx 2 \times 10^{11} \frac{N}{m^2}$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{AY}{mL_0}}$$

$$T = 2\pi \sqrt{\frac{mL_0}{AY}}$$

Steel wire: $L_0 \approx 1\,m$

$d \approx 1\,mm$

$$A = \frac{\pi d^2}{4} \approx 0.8 \times 10^{-6}\ m^2$$

$$k = \frac{AY}{L_0} = \frac{0.8 \times 10^{-6} \times 2 \times 10^{11}}{1} \approx 1.6 \times 10^5$$

$2\times10^{11}\ N/m^2$

$m = 1\ kg$

$\omega^2 = \frac{k}{m} = 16\times10^4\ s^{-2}$

$\omega \approx 400\ s^{-1}$

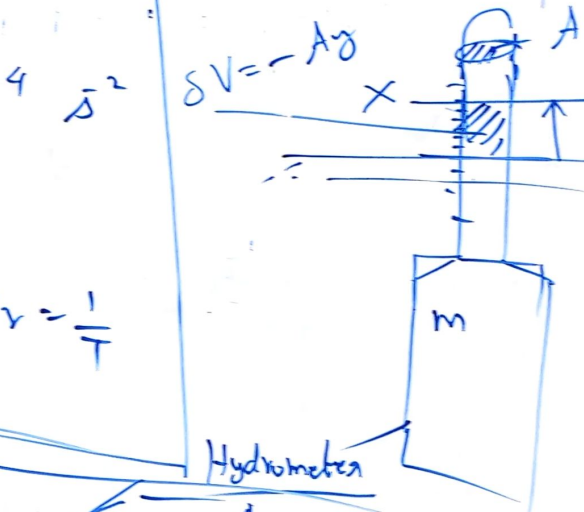
$\nu = \frac{\omega}{2\pi} \simeq 60\ Hz = \frac{1}{T}$

$mm$ —

$\times10^{-6}\ m^2$

$\times2\times10^{11}$

$= 1.1\times10^5\ N/m$




$\delta V = -Ay$

$\rho$ = density of liq.

## Ships

$m \approx A h \rho_{ship}$

$\omega = \sqrt{\frac{g}{h}}, \quad T = 2\pi\sqrt{\frac{h}{g}}$

In Equilibrium →

$m\frac{d^2y}{dt^2} = -mg + g\rho V_{liq}^{\circ}$ (1)

$= 0.$

↑ Vol. of liq. displaced in equil.

$\Rightarrow mg = g\rho V_l^{\circ} \quad (y=0)$

$y \neq 0: \quad V_{liq} = V_{liq}^{\circ} - Ay$ (2)

$y=0 \to$ (1)

$m\frac{d^2y}{dt^2} = -mg + g\rho V_{liq}$

$\boxed{\frac{d^2y}{dt^2} = -\frac{g\rho A}{m}y}$

$\omega = \sqrt{\frac{g\rho A}{m}}$